elements below, is a nonmetal

A) K (atomic no. = 19)
B) Mg (atomic no. = 12)
C) Li (atomic no. = 3)
D) Cl (atomic no. = 17)
E) Ge (atomic no. = 32)

Select one:

- A
- B
- C
- D
- E

How many electrons are there in the valence shell of the Si ion?

(atomic number =14)

A. 2

B. 8

C. 7

D. 6

E. 4

Select one:

- A
- B
- C
- D
- E

._____ (n) of the first p- orbital is

A. 1
B. 2
C. 3
D. 4
E. 0

إختر أحد الخيارات:

A
B
C
D

The molar mass of compound S_4O_x = 224 g/mol

Atomic weights: S = 32, 0 =16 g/mol

?What is the value x

A. 3

B. 5

C. 7

D. 4

E. 6

إختر أحد الخيارات:

A

B

C

1s 2s 2p 3s 3p 4s 3d is a ground state configuration for a

A. Transition element
B. Representative element
C. Halogen
D. Inner transition element
E. Nonmetal element

إختر أحد الخيارات:

A
B
C
D
E

What is the atomic number for an element that belongs to period 5, group 2B in the periodic table?

A. 47

B. 42

C. 41

D. 80

E. 48

إختر أحد الخيارات:

A

B

C

D

Which element has the lowest electron affinity (EA) for

$_{9}F$, $_{3}Li$, $_{10}Ne$

A. F

B. Li

C. Ne

إختر أحد الخيارات:

A

B

C

(Atomic weight of Ca = 40 g/mol)

A. 1.0 mole

B. 0.5 mole

C. 2.5 mole

D. 2.0 mole

E. 1.5 mole

إختر أحد الخيارات:

A

B

C

D

E

The element X (atomic number = 30) is a_____ element with unpaired electrons _____

A. paramagnetic , 3

B. paramagnetic , 2

C. paramagnetic, 6

D. diamagnetic , 2

E. diamagnetic , 0

إختر أحد الخيارات:

A

B

Which of the elements below, is a nonmetal

A) K (atomic no. = 19)
B) Mg (atomic no. = 12)
C) Li (atomic no. = 3)
D) Si (atomic no. = 14)
E) S (atomic no. = 16)

إختر أحد الخيارات:

- A
- B
- C
- D
- E

of atoms) are found

in 7.5 mole of $K_2Cr_2O_7$

Avogadro's number = 6.022×10^{23}

A. 2.3×10^{25}

B. 7×10^{25}

C. 5×10^{25}

D. 1.4×10^{26}

E. 3.6×10^{24}

إختر أحد الخيارات:

A

describes the energy of an orbital is:

A. Principle quantum number

B. Magnetic quantum number

C. Secondary (angular momentum) quantum number

D. Spin quantum number

E. None of the above

إختر أحد الخيارات:

A

B

C

D

The first ionization energy for elements _____ from LEFT to RIGHT through a period and _____ from BOTTOM to the TOP through a group in the periodic table.

A. increase, increase

B. increase, decrease

C. decrease, increase

D. decrease, decrease

E. is constant, decrease

Select one:

- A
- B
- C
- D
- E

The quantum number that describes the shape of an orbital is:

A. Principle quantum number
B. Magnetic quantum number
C. Secondary (angular momentum) quantum number
D. Spin quantum number
E. None of the above

Select one:

- A
- B
- C
- D
- E

$$__NH_{3\,(g)} + __O_{2(g)} \rightarrow __N_{2(g)} + __H_2O_{(l)}$$

The coefficients in the balanced equation :(respectively) are

A. 4,3,2,6
B. 2,1,2,3
C. 1,3,1,3
D. 2,3,2,3
E. 3,4,6,2

إختر أحد الخيارات:

A
B
C
D

Calculate the mass percentage of nitrogen (N) in of $Al(NO_3)_2$

(Atomic weights: Al= 27 , N=14, O =16 g/mol)

A. 17.1%

B. 19.7%

C. 58.5%

D. 67.6%

E. 12.2%

Select one:

- A
- B
- C
- D
- E

the secondary quantum number (l) is:

A. 1

B. 3

C. 4

D. 0

E. 2

Select one:

- A
- B
- C
- D
- E

$2C_3H_7OH + 9O_2 \rightarrow 6CO_2 + 8H_2O$

If 12 g C_3H_7OH (60 g/mol) are allowed to react with 40 g O_2 (32g/mol)

The mass of CO_2 (44g/mol) produced in grams is:

A. 13.75
B. 18.33
C. 22.91
D. 26.4
E. 17.6

Select one:

- A
- B
- C
- D
- E

The molar mass of compound S_2O_x = 112 g/mol

Atomic weights: S = 32, O =16 g/mol

What is the value x?

A. 3

B. 5

C. 7

D. 4

E. 6

Select one:

- A
- B
- C
- D
- E

How many moles is 100 g of Ca?
(Atomic weight of Ca = 40 g/mol)

A. 1.0 mole
B. 0.5 mole
C. 2.5 mole
D. 2.0 mole
E. 1.5 mole

Select one:

- A
- B
- C
- D
- E

How many atoms (total number of atoms) are found in 0.5 mole of $K_2S_4O_6$

Avogadro's number = 6.022×10^{23}

A. 2.3×10^{25}

B. 7×10^{25}

C. 5×10^{25}

D. 1.4×10^{26}

E. 3.6×10^{24}

Select one:

- A
- B
- C
- D
- E

Which element has the lowest IE_1 in period 2 ?

A. Ne

B. Ar

C. Li

D. Na

Select one:

- A
- B
- C
- D

Aluminum reacts with oxygen to produce aluminum oxide.

$4\ Al_{(s)} + 3\ O_{2(g)} \rightarrow 2\ Al_2O_{3(s)}$

If 10.0 moles of Al react with excess O_2, how many moles of Al_2O_3 can be formed?

A. 1.5 mol
B. 2.0 mol
C. 2.5 mol
D. 5.0 mol
E. 3.0 mol

Select one:

- A
- B
- C
- D
- E

ground state atom has the

quantum number set (n, l, m_l, m_s) equal to $(2,0,0, +½)$

(Atomic no. of N =7)

A. first electron

B. fourth electron

C. fifth electron

D. sixth electron

E. seventh electron

Select one

- A
- B
- C
- D
- E

Which electron in the nitrogen ground state atom has the quantum number set (n, l, m_l, m_s) equal to $(2,1,0, +\frac{1}{2})$ (Atomic no. of N = 7)

A. first electron

B. third electron

C. fifth electron

D. sixth electron

E. seventh electron

Select one:

- A
- B
- C
- D

If 0.6 mol ZnS are allowed to react with excess amount of O_2

Calculate percentage yield of the reaction given that 10 g of SO_2 (64g/mol) is actually produced :

A. 86.8 %
B. 60 %
C. 43.4%
D. 26 %
E. 30 %

Select one:

- A
- B
- C
- D